

EMIPredict AI

EMI Eligibility and Maximum EMI Prediction System Using Machine Learning

Mizaru John

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1 Introduction

Financial institutions evaluate thousands of loan applications every day. Traditional loan assessment methods rely heavily on manual analysis of customer income, expenses, credit history, and repayment capacity.

Manual evaluation introduces several challenges:

- Slow decision making
- Human bias
- Inconsistent approvals
- Increased risk of loan defaults

To address these issues, this project develops a Machine Learning based system capable of:

1. Predicting EMI Eligibility
2. Predicting Maximum Affordable EMI
3. Assessing Financial Risk

The solution combines Machine Learning, MLflow, and Streamlit to create an end-to-end intelligent lending support system.

2 Problem Statement

The objective is to determine:

- Whether a customer is eligible for EMI approval.
- Whether the customer falls into a High Risk category.
- The maximum EMI amount the customer can safely afford.

The system should provide accurate predictions while minimizing lending risk.

3 Dataset Overview

The dataset contains customer demographic, employment, financial, and loan-related information.

Attribute	Value
Records	404,800
Features	35
Missing Values Before Cleaning	15,986
Missing Values After Cleaning	0
Classification Target	emi_eligibility
Regression Target	max_monthly_emi

Key features include:

- Age
- Gender
- Education
- Monthly Salary
- Employment Type
- Credit Score
- Bank Balance
- Emergency Fund
- Requested Loan Amount
- Requested Tenure

4 Data Cleaning

Raw datasets often contain missing values, inconsistent formats, and incorrect datatypes. Data cleaning was performed to ensure model reliability.

4.1 Missing Value Treatment

Numerical variables:

- Filled using median values.

Categorical variables:

- Filled using mode values.

4.2 Datatype Correction

Several numerical columns were incorrectly stored as object datatypes.

Examples:

- `monthly_salary`
- `bank_balance`
- `emergency_fund`
- `credit_score`

These columns were converted into numeric format before analysis.

4.3 Why Data Cleaning Was Necessary

Machine Learning algorithms require complete and consistent datasets.

Without cleaning:

- Model accuracy decreases.
- Training errors occur.
- Business insights become unreliable.

5 Exploratory Data Analysis

EDA was performed to understand customer behavior and identify patterns.

5.1 Eligibility Distribution

The dataset was highly imbalanced.

Class	Percentage
Not Eligible	77.29%
Eligible	18.39%
High Risk	4.32%

5.2 Key Findings

1. Higher credit scores increased eligibility.
2. Higher monthly salary increased EMI affordability.
3. Bank balance positively influenced repayment capacity.
4. Emergency funds reduced financial risk.
5. Existing loans negatively impacted eligibility.

5.3 Why EDA Was Performed

EDA helps:

- Understand relationships among variables.
- Detect imbalance.
- Identify influential predictors.
- Guide feature engineering decisions.

6 Feature Engineering

Feature Engineering transforms raw data into meaningful variables.

This step significantly improves model performance.

6.1 Features Created

1. Total Expenses
2. Debt To Income Ratio
3. Expense Ratio
4. Savings Ratio
5. Financial Cushion
6. Employment Stability
7. Loan Burden Ratio
8. Available Income

6.2 Why Feature Engineering Was Necessary

Raw variables often fail to capture customer financial health.

For example:

$$DebtToIncomeRatio = \frac{CurrentEMI}{MonthlySalary} \quad (1)$$

This ratio provides a stronger indicator of affordability than either variable alone. Similarly:

$$AvailableIncome = MonthlySalary - Expenses - CurrentEMI \quad (2)$$

This feature directly measures repayment capability.

7 Classification Modeling

The classification target was:

emi_eligibility

Classes:

- Eligible
- High Risk
- Not Eligible

7.1 Models Evaluated

1. Logistic Regression
2. Random Forest Classifier
3. XGBoost Classifier

7.2 Why These Models Were Chosen

Logistic Regression

- Simple baseline model.
- Easy interpretation.

Random Forest

- Handles nonlinear relationships.
- Robust against overfitting.

XGBoost

- State-of-the-art gradient boosting algorithm.
- Excellent performance on tabular datasets.
- Handles complex interactions.

7.3 Results

Model	Accuracy	F1 Score
Logistic Regression	79.44%	84.38%
Random Forest	87.82%	90.20%
XGBoost	96.79%	96.30%

7.4 Best Model

XGBoost Classifier achieved the highest overall performance and was selected for deployment.

8 Regression Modeling

Target variable:

max_monthly_emi

8.1 Models Evaluated

1. Linear Regression
2. Random Forest Regressor
3. XGBoost Regressor

8.2 Evaluation Metrics

- RMSE
- MAE
- MAPE
- R^2

8.3 Results

Model	RMSE	MAE	R^2
Linear Regression	4092.60	2933.61	0.7163
Random Forest	991.11	298.44	0.9834
XGBoost	674.06	226.30	0.9923

8.4 Best Model

XGBoost Regressor achieved the lowest prediction error and highest R^2 score.

9 MLflow Integration

MLflow was integrated to manage experiments.

9.1 Purpose

- Track experiments
- Compare models
- Store metrics
- Store trained models

9.2 Experiments Logged

- Best_Classifier_XGBoost
- Best_Regressor_XGBoost

MLflow improved reproducibility and model management.

10 Streamlit Deployment

A multi-page Streamlit application was developed.

Pages included:

1. Home Page
2. EDA Dashboard
3. EMI Eligibility Prediction
4. Maximum EMI Prediction
5. Model Performance Dashboard

The application enables real-time predictions through a user-friendly interface.

11 Business Impact

The proposed solution offers:

- Faster loan approval decisions
- Improved risk assessment
- Reduced default probability
- Consistent decision making
- Enhanced customer experience

12 Conclusion

This project successfully developed an end-to-end Machine Learning solution for EMI eligibility classification and EMI affordability prediction.

The XGBoost Classifier achieved an accuracy of 96.79%, while the XGBoost Regressor achieved an R^2 score of 0.9923.

The integration of Feature Engineering, MLflow, and Streamlit resulted in a production-ready financial decision support system.

13 Future Scope

- Real-time API deployment
- Cloud deployment on AWS
- Integration with credit bureau services
- Explainable AI using SHAP
- Deep Learning based risk assessment